



MAINTENANCE

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UNIVERSITY SYSTEM

ELEVATORS

WHAT IS AN ELEVATOR?

DEFINITION

- Permanently installed.
- Car intended for transporting people and objects.
- Suspended by cables, chains, pulleys, or pistons.
- Cable inclination relative to vertical of less than 15°.

HOW AN ELEVATOR WORKS

ELEVATORS ARE VERTICAL TRANSPORTATION SYSTEMS THAT ALLOW PEOPLE AND GOODS TO MOVE BETWEEN DIFFERENT LEVELS OF A BUILDING. THEY ARE AN ESSENTIAL PART OF MODERN LIFE, AND THEIR OPERATION RELIES ON A COMBINATION OF MECHANICS, ELECTRONICS, AND HYDRAULICS.



COMPONENTS OF AN ELEVATOR

MOTOR

THE MOTOR IS THE HEART OF THE ELEVATOR. IT IS RESPONSIBLE FOR DRIVING THE CABLE THAT RAISES AND LOWERS THE CAR.

CAR

THE CAR IS WHERE PASSENGERS RIDE. IT IS DESIGNED TO PROVIDE A SAFE AND COMFORTABLE TRIP.

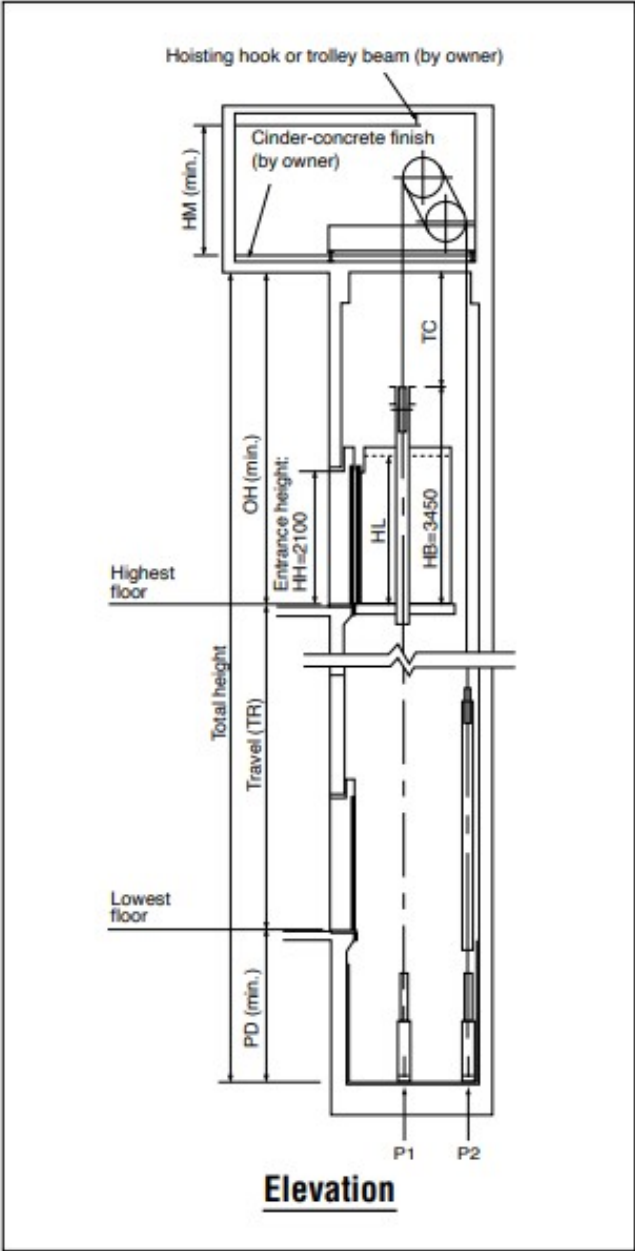
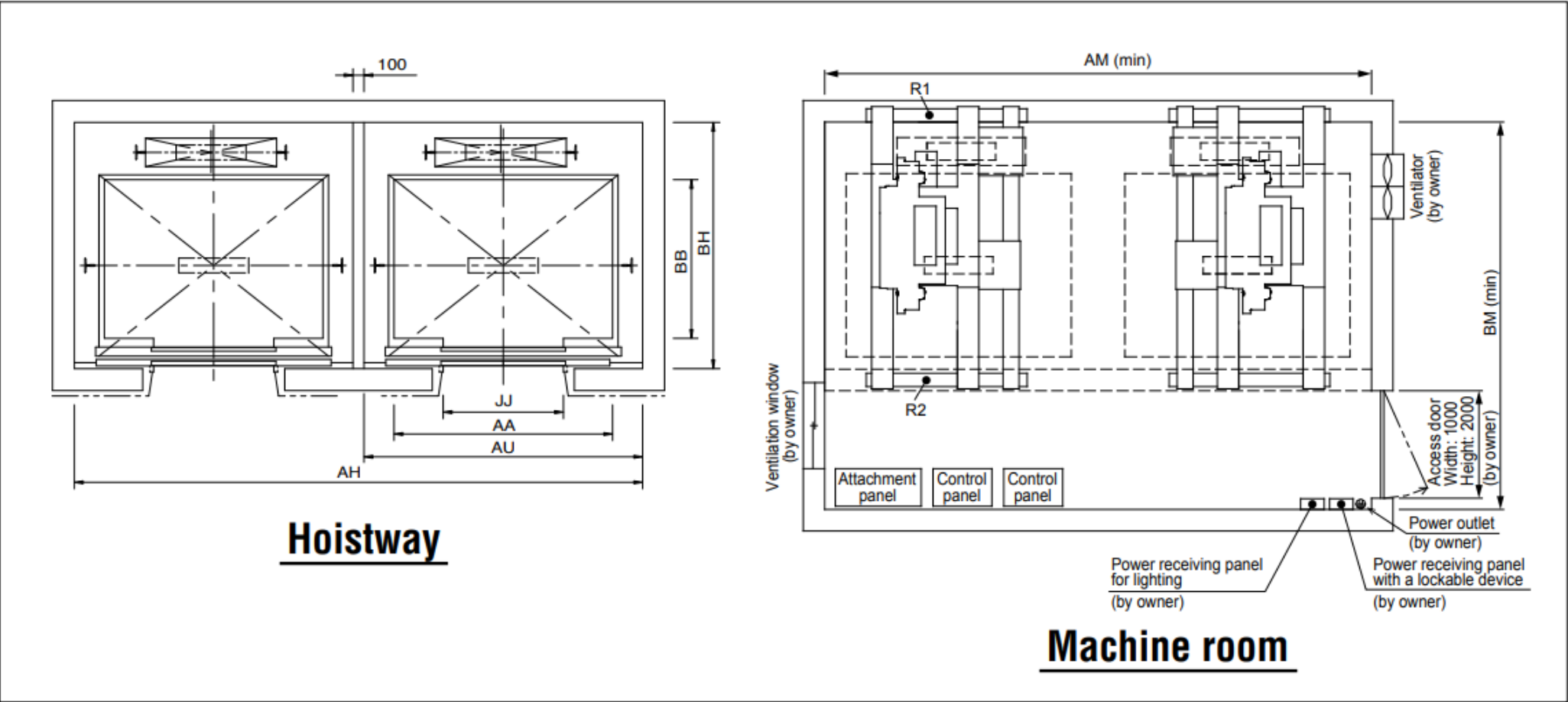
COUNTERWEIGHTS

COUNTERWEIGHTS ARE USED TO BALANCE THE WEIGHT OF THE CAR AND PASSENGERS, WHICH REDUCES THE AMOUNT OF ENERGY NEEDED TO OPERATE THE ELEVATOR.

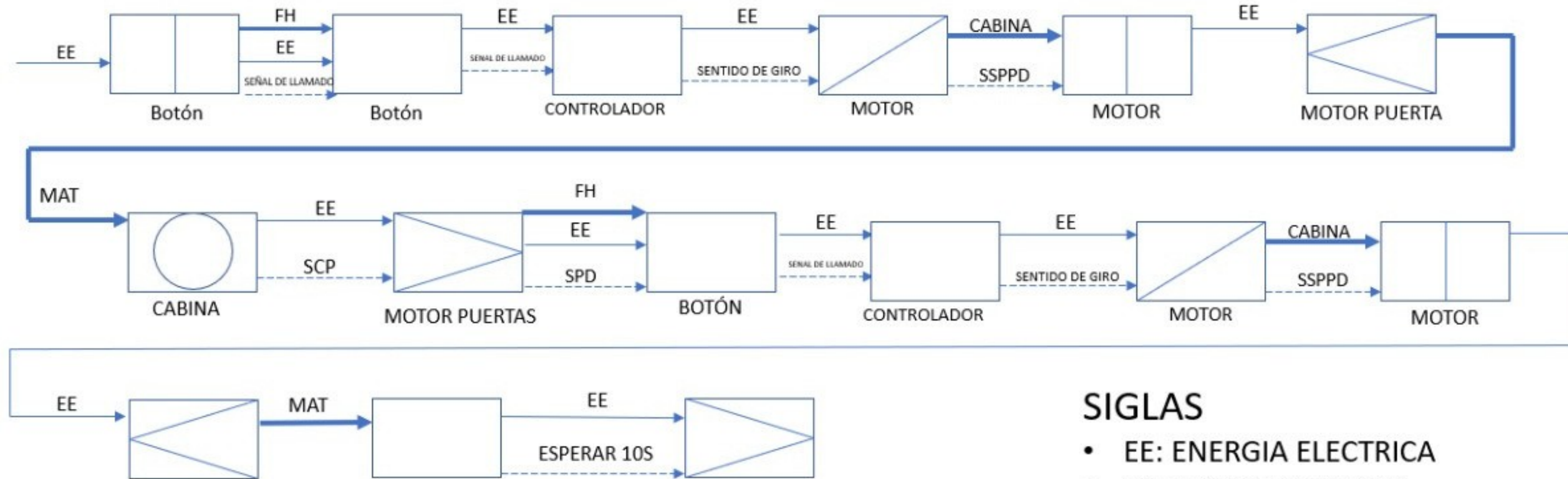


LAYOUT

Layout example of in-line arrangement



ELEVATOR FUNCTIONAL STRUCTURE



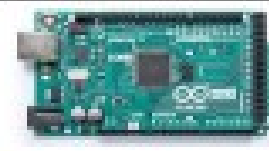
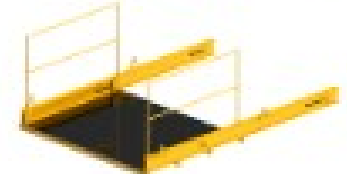
Verbos de Pahl & Beitz

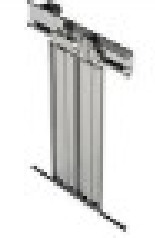
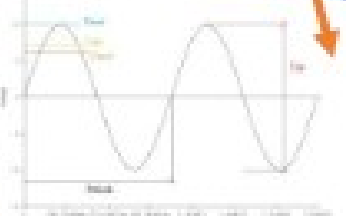
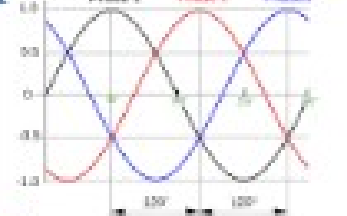
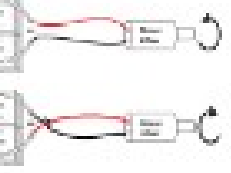


SIGLAS

- EE: ENERGIA ELECTRICA
- FH: FUERZA HUMANA
- SSPPD: SEÑAL SENSOR PISO DESEADO
- MAT: MATERIAL A TRANSPORTAR
- SCP: SEÑAL CIERRE PUERTAS

MORPHOLOGICAL MATRIX

Funciones parciales	Portador de función 1	Portador de función 2	Portador de función 3
Sistema de entrega de potencia			-
Sistema de control			
Sistema de transmisión			
Sistema móvil			

Puertas			
Alimentación de energía			
Sistema de frenado			
Mecanismo elevador			

Instalación y Mantenimiento del Elevador

1

Planificación

La instalación comienza con la planificación del diseño del elevador, teniendo en cuenta las características del edificio.

2

Instalación del Pozo

Se excava el pozo y se instala el sistema de guía para el movimiento del elevador.

3

Instalación de la Cabina

La cabina se instala en el pozo y se conecta a los sistemas de control y seguridad.

4

Pruebas y Puesta en Marcha

Antes de la puesta en marcha, se realizan pruebas exhaustivas para garantizar la seguridad y el correcto funcionamiento del elevador.



ELEVATOR CONTROL SYSTEMS

SENSORS

SENSORS DETECT THE POSITION OF THE CAR, PASSENGER MOVEMENT, AND OTHER FACTORS IMPORTANT TO THE ELEVATOR'S OPERATION.

CONTROLLER

THE CONTROLLER RECEIVES INFORMATION FROM THE SENSORS AND USES IT TO CONTROL THE ELEVATOR'S MOVEMENT.

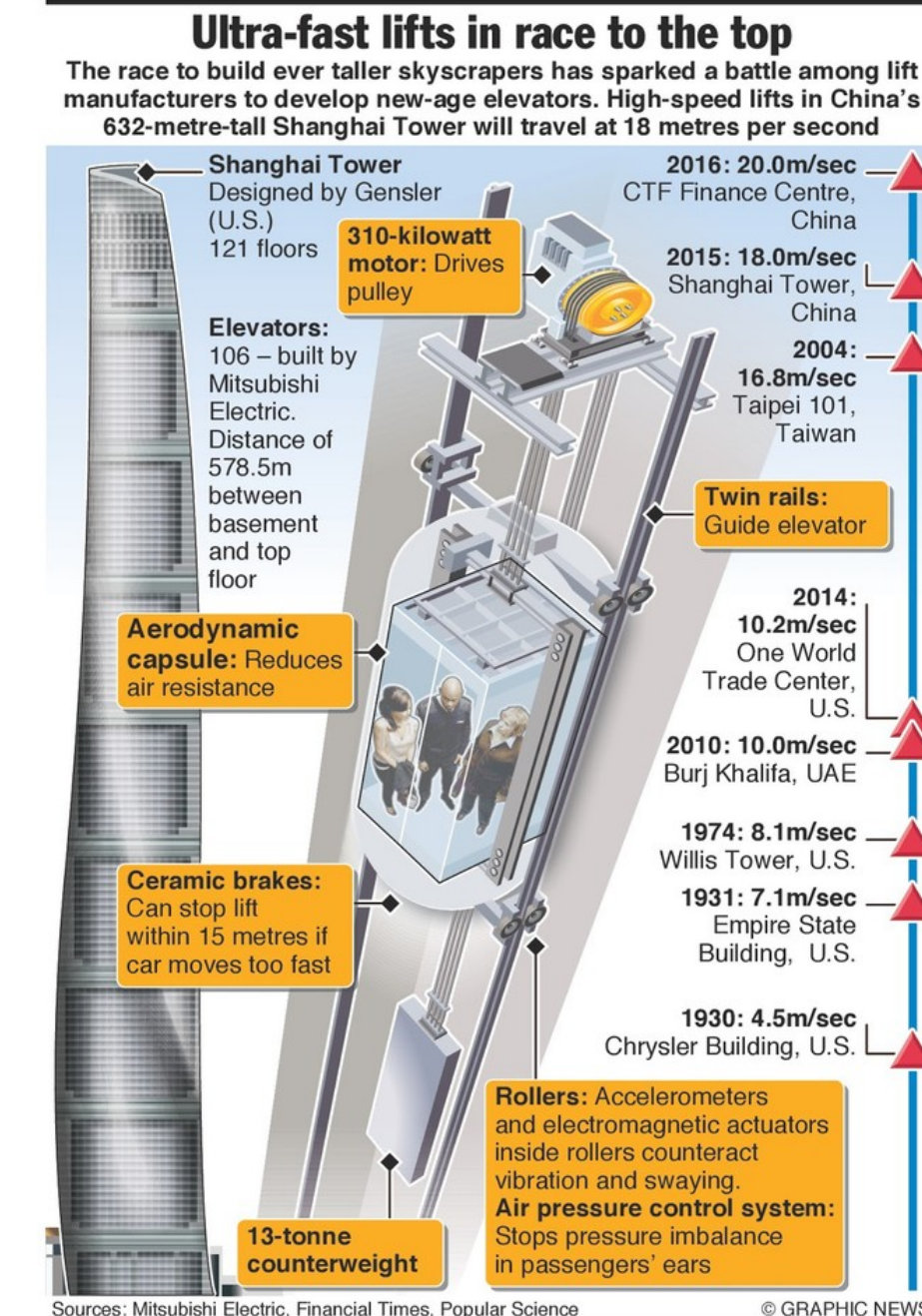
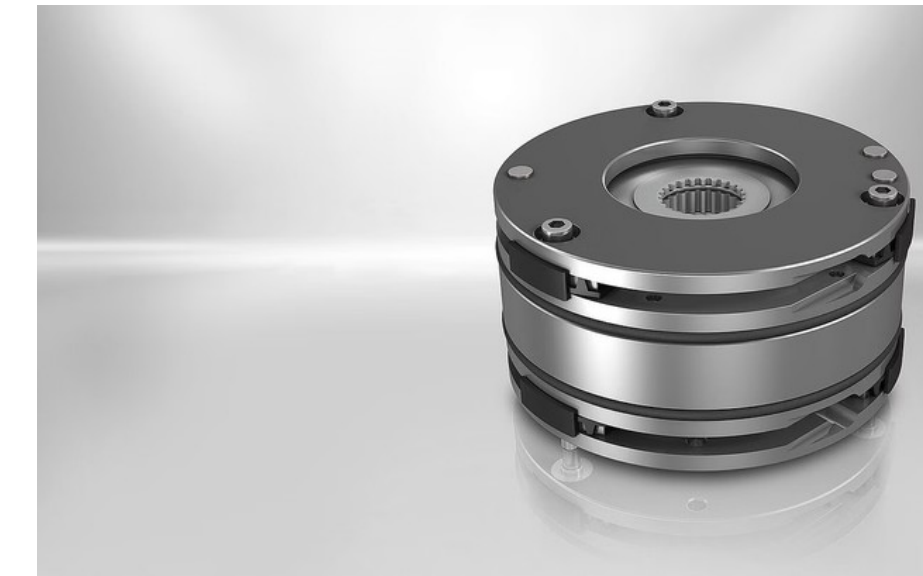
USER INTERFACE

THE USER INTERFACE ALLOWS PASSENGERS TO SELECT THE FLOORS THEY WISH TO VISIT.



Características de Seguridad del Elevador

- 1 Freno de Seguridad**
Los frenos de seguridad son un sistema de seguridad crucial que evita que la cabina caiga en caso de fallo del sistema principal.
- 2 Puertas Automáticas**
Las puertas automáticas evitan que los pasajeros entren o salgan del elevador cuando está en movimiento.
- 3 Iluminación de Emergencia**
La iluminación de emergencia proporciona luz en caso de corte de energía, asegurando la seguridad de los pasajeros.
- 4 Botón de Alarma**
El botón de alarma permite a los pasajeros comunicarse con el personal de mantenimiento en caso de emergencia.



NORMA TÉCNICA COLOMBIANA

**NTC
5926-1**

2012-09-24

**CRITERIOS PARA LAS INSPECCIONES DE
ASCENSORES, ESCALERAS MECÁNICAS,
ANDENES MÓVILES Y PUERTAS ELÉCTRICAS.
PARTE 1: ASCENSORES ELECTROMECAÑICOS E
HIDRÁULICOS**

E: CRITERIA FOR INSPECTIONS OF ELEVATORS,
ESCALATORS, MOVING WALKS AND ELECTRIC GATES.
PART 1: ELECTROMECHANICAL AND HYDRAULIC LIFTS

STANDARD

The standard that governs
elevators, escalators,
moving walkways, and
power-operated doors.



NTC – 5926 – 1

For elevators, this is part
one of the standard, which
establishes the safety and
inspection requirements for
these systems.

**NORMA TÉCNICA
COLOMBIANA**

**NTC
4349**

2017-03-22

**REGLAS DE SEGURIDAD PARA LA
CONSTRUCCIÓN E INSTALACIÓN DE
ASCENSORES.
APLICACIONES PARTICULARES PARA
ASCENSORES DE PASAJEROS Y DE
PASAJEROS Y CARGAS. ACCESIBILIDAD A LOS
ASCENSORES DE PERSONAS, INCLUYENDO
PERSONAS CON DISCAPACIDAD**

E: SAFETY RULES FOR THE CONSTRUCTION AND
INSTALLATIONS OF LIFTS. PARTICULAR APPLICATIONS
FOR PASSENGER LIFTS. ACCESSIBILITY TO LIFTS FOR
PERSONS INCLUDING PERSONS WITH DESABILITY.

CORRESPONDENCIA: esta norma es una adopción
modificada (MOD) de la norma EN
81-70:2003.

DESCRIPTORES: ascensor; discapacidad; dimensiones;
cabina; accesibilidad.

STANDARD

The standard that governs
passenger and freight
elevators.



NTC – 4349

which establishes safety
requirements for the
construction and
installation of elevators.



COMPETENT PERSON

A person with demonstrated ability to apply knowledge and skills to a given work activity.



INSPECTION

Examination of a product's design and the product itself, and determination of its conformity with specific requirements or on the basis of professional judgment



EXCEPTIONS TO THE STANDARD



SHIPS

Equipment used to outfit ships and offshore exploration facilities.



MINES

Elevators installed in mines or used to handle radioactive material.



CONSTRUCTION SITES

Rack-and-pinion elevators, strip lifts, and hoists used on building construction sites.

ELEVATOR BRANDS AT THE UNIVERSITY



ESTILO



SCHINDLER



MITSUBISHI

RELEVANT DATA

ESTILO

2-YEAR WARRANTY
COLOMBIAN COMPANY

PERSONALIZED
MAINTENANCE SERVICE
WITH LESS PREDICTIVE
TECHNOLOGY

COMPLIES WITH
RELEVANT LOCAL AND
INTERNATIONAL
REGULATIONS

SCHINDLER

PREDICTIVE
MAINTENANCE WITH
ADVANCED
TECHNOLOGY.
FOCUS ON USER
INTERFACE AND
EFFICIENCY

STRICT COMPLIANCE
WITH INTERNATIONAL
STANDARDS SUCH AS
ISO AND EN

MITSUBISHI

HIGHLY
TECHNOLOGICAL
MAINTENANCE
WITH REMOTE
MONITORING.

RIGOROUS
COMPLIANCE WITH
ISO AND JIS
STANDARDS, WITH A
FOCUS ON
EFFICIENCY AND
SAFETY.



WITH WHAT WE
HAVE...

Viable ways to reduce costs



1-Adopt a foreign standard that is more rigorous than the Colombian one.

2.-Modernize/upgrade the model of some elevators, pursuing the energy efficiency these companies promise with each updated model (and making sure spare parts are available).

3-Reduce the number of elevators at the university.





UNIVERSIDAD
EAFIT®

THANK YOU!

**THANK YOU VERY MUCH FOR
YOUR ATTENTION!**



Questions?